

Kundai Farai Sachikonye

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Bioinformatics / cheminformatics engineer: semantic data standards (OWL/RDF), ontologies and DSLs, agentic and closed-loop research tooling — applied to enzyme catalysis and biocatalytic data.

Profile

Computational scientist and software engineer working at the intersection of **biocatalysis, semantic data, and research automation**. I design ontologies and domain-specific languages, build FAIR-aligned data tooling, and write the agentic infrastructure that drives closed-loop experimentation — and I bring a working enzyme-kinetics / catalysis background to ground the data models in the chemistry they describe. Everything I build is deployed, documented, and open-source.

Fit to NFDI4Cat

Semantic data & FAIR	Design of ontologies, taxonomies and domain-specific languages; OWL / RDF knowledge representation; FAIR-data-aligned schemas for scientific exchange
(Bio)catalysis domain	Enzyme kinetics and catalytic-mechanism modelling (e.g. carbonic anhydrase turnover, transition-state vs. geometric frameworks); mass-spectrometry data
Agentic & closed-loop	purpose — an agent-queryable knowledge layer over scientific corpora; autonomous, iterative experiment / measurement pipelines
Engineering & DevOps	Python, JavaScript/TypeScript, Rust; Docker (Kubernetes); Git, agentic programming; relational / scientific databases

Technical Skills

Languages	Python, JavaScript / TypeScript, Rust, Java, PHP; SQL
Semantic web	OWL, RDF; ontology / taxonomy design; domain-specific languages and their compilers (lexer / parser / codegen)
ML / AI	PyTorch; agentic / multi-agent systems; foundation-model adaptation (LoRA); RAG and reasoning pipelines; logic / rule-based reasoning
Infrastructure	Docker, Kubernetes; Ray, Dask, SLURM; AWS, Firebase; large-scale (>100 GB) pipelines; Git
Developer tooling	5 VSCode extensions including a language server (LSP); reproducible CLI tooling

Experience

2021–present	Independent Computational Researcher & Software Engineer, Bitspark GmbH Built compilers / domain-specific languages, ontology-backed knowledge tooling, GPU pipelines and agentic AI systems — all deployed and open-source. Authored a research corpus on enzyme catalysis, spectroscopy and (bio)catalytic data representation. Rust, Python / PyTorch, TypeScript.
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- 2019–2021 **Computational Science — Bioinformatics, HPC & Teaching**, *Technische Universität München*, Munich
 Doctoral research in computational lipidomics; large-scale data analysis and ML pipelines at HPC scale (PyTorch, SLURM). Built and curated lipidomics databases. Taught at Master's level and supervised student work.
- 2017–2018 **Data Pipelines**, *Universitätsklinikum Hamburg-Eppendorf*, Hamburg
 Automated, reproducible high-throughput data-processing pipelines.

Selected Projects

- purpose Rust CLI (Cargo workspace): an agent-queryable context layer over scientific project corpora — the closed-loop / agentic-programming pattern the NFDI4Cat role calls for. github.com/fullscreen-triangle/purpose
- kwasa-kwasa A polyglot programming language with its own compiler and a suite of 5 VSCode extensions (incl. a language server); DSL / semantic-runtime design. Rust + TypeScript. github.com/fullscreen-triangle/kwasa-kwasa
- Enzyme-catalysis research First-principles treatment of enzyme turnover and catalytic mechanism (carbonic anhydrase), with platform-invariant mass-spectrometry data models — domain grounding for biocatalytic data standards.
- lavoisier Mass-spectrometry analysis framework: platform-invariant featurization and reproducible scientific pipelines (Python). github.com/fullscreen-triangle/lavoisier

Education

- 2018–2019 **MSc Computational Biology**, *Universität Tübingen*
- 2014–2017 **MSc Pharmaceutical Biotechnology**, *HAW Hamburg*
- 2011–2014 **BSc Biochemistry and Cell Biology**, *Jacobs University Bremen*
Additional: doctoral research, computational lipidomics, TUM (not completed).

Languages

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| English | Native / Fluent (C2) | <i>required for the role</i> |
| German | Conversational | <i>resident in Germany since 2011</i> |

Work Authorization

Permanent residence in Germany — unrestricted work authorization, no sponsorship required.